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Dynamic Dual Zone Smart Completions for Problematic Wells: A Case Study from the Maturing Asset in UAE / Abu Dhabi

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Abstract

To maximize production across two distinct reservoirs, ADNOC onshore initiated the deployment of its first smart completion in a high-risk integrity well. This study outlines the planning and execution phases, highlighting the successful use of a light workover rig to install a completion equipped with Interval Control Valves (ICVs), Permanent Downhole Gauges (PDHGs), and Gas Lift Mandrels (GLMs) into a 7" casing tie-back. The solution enhanced zonal isolation, improved inflow performance, and enabled real-time inflow/outflow control, preparing the well for sustained natural flow without further intervention.

Comprehensive engineering simulations, risk assessments, and system integration tests were conducted to validate tool functionality and the deployment of multiple control lines through the packer, ICVs, and PDHGs. Collaboration between ADNOC onshore and SLB ensured site readiness and special equipment provisioning through multiple rig visits and continuous evaluations.

The wellbore was meticulously cleaned to remove debris that could obstruct the installation of smart completion assemblies. Challenges encountered at various depths were resolved using scrapers, magnets, and junk baskets. After reaching target depth, wireline operations confirmed packer setting depths. These cleaning techniques offer valuable reference for similar operations across the UAE and internationally.

The smart completion enabled real-time control, production optimization, improved zonal isolation, and inflow control via a dual sensor monitoring which can result in higher production rates and improved efficiency. By monitoring downhole conditions in real-time, operators can quickly detect and address any production issues, reducing downtime and increasing well availability. It also eliminated the need for a replacement well, rigless workovers for selective production, wireline gradient surveys, and PLTs for zonal allocation—delivering significant operational efficiencies and cost savings.

This case study demonstrates the successful deployment of smart completion technology in a challenging environment, showcasing the benefits of real-time monitoring and control for enhanced reservoir management and hydrocarbon recovery. It provides actionable insights and best practices for future smart completion projects in similar scenarios.

Field Background and Operational Context

A well located in a mature onshore field within the South-East Asset marked a significant milestone in deployment of smart completion technology in the region. Originally drilled in June 1994 as a dual zone vertical oil producer targeting two distinct reservoir layers, the well faced progressive integrity challenges leading to its shutdown in August 2021. These included SAP "B" (casing to casing communication), downhole flow assurance issues, and mechanical communication between strings, leading to water holdup and loss of reservoir accessibility. Due to the integrity concern it was necessary to cure SAP "B" by conducting section mill operation to stop the communication in the annulus space between 9-5/8"and 13-3/8"casing followed by cemented 7"tie back strategy potentially abandoning one of the reservoir targets due to well integrity and barrier challenges. Recognizing the strategic value of restoring dual-zone production, the operator selected this well as a pilot candidate for its first smart completion in this field. The workover was executed in December 2023, with the objective of mitigating integrity, corrosion, and flow assurance challenges through intelligent completion solutions. The smart completion system, incorporated dual hydraulic surface-controlled Interval Control Valves (ICVs) with six adjustable choke positions and two on/off settings. This configuration enabled selective and commingled production from both reservoir zones, supported by permanent downhole gauges for real-time monitoring of pressure and temperature across both zones. Commissioning activities spanned from February to August 2024 and included acid stimulation, nitrogen lift, installation of digital wellhead gauge, and zone-specific testing. The results were transformative: production resumed from the upper reservoir at ~1000 bpd with <2% water cut, while the lower reservoir contributed an additional 300–400 barrels per day through commingled flow. This eliminated the need for a replacement well and delivered an estimated CAPEX saving of $4.8 million. To ensure long-term performance, the operator implemented a tailored Reservoir Monitoring Plan (RMP) including monthly production tests, annual production logging tests (PLTs), and continuous gradient monitoring via PDHGs. An internally developed physics-based workflow was used for zonal allocation, validated against well test and logging data. This case sets a benchmark for intelligent field rehabilitation, demonstrating how smart completions can unlock previously inaccessible reserves, extend well life, and support broader digitalization and recovery optimization initiatives.

Workover Design & Execution

The workover operation was technically demanding aimed at restoring well integrity and optimizing production through the deployment of a smart completion system. The key objectives included enabling selective zonal control, enhancing downhole surveillance, and ensuring flexibility for future artificial lift and commingled production scenarios.

Planning and Rig Mobilization

The campaign began with a detailed planning phase focused on reactivating two reservoir zones that had been shut in due to severe integrity issues, including packer failures, tubing communication, and debris accumulation. A cross-functional team developed a strategy to convert the well into a dual-zone producer using smart completion technology. Risk assessments were conducted for debris management, hydraulic/electrical integrity, and concentric equipment alignment while sub-assembly make up to fit within the completion architecture.

Rig mobilization involved relocating equipment with 51 planned loads including chemical packages. Civil equipment is supported to facilitate site preparation under full HSE compliance. Rig acceptance was confirmed with a high inspection score, and pre-operation alignment meetings ensured readiness, particularly for handling surface equipment in congested areas.

Cleanout Operations and Debris Management

Initial operations focused on well control assurance, BOP testing, and cleanout runs. The wellbore contained metallic debris from previous interventions. A combination of scrapers, magnets, junk baskets, and high-viscosity pills was used to clear obstructions.

- **Multiple Cleanout Runs:** Five runs were performed due to persistent debris, with magnet assemblies achieving full coverage and recovering over 5.7 kg of metallic material.
- **Operational Flexibility:** The team adapted by modifying assemblies with junk subs and additional magnets, ensuring a clean wellbore for subsequent completion stages.

Safety protocols were rigorously followed, including scheduled safety time-outs and emergency drills.

Section Milling and 7″ Production Liner Tieback

A critical component of the workover was the removal of legacy barriers and the establishment of a new conduit for smart completion. Section milling was performed at multiple depths using an 8½″ section mill assembly, dressed with bumper subs, fishing jars, and heavy-weight drill pipe. Milling intervals included 2535 ft and 495 ft, targeting the 9⅝" casing to create a clean window for liner tieback.

Each milling run was followed by circulation and junk recovery. Torque and drag were closely monitored, and jarring was used to free the string when encountering tight spots. The operation required precise spacing out and tool handling to avoid equipment damage.

Following successful milling, a 7″ production liner was run from the milled window to surface. The liner included a tieback seal assembly and fluted casing hanger. It was pressure-tested to 500 psi and verified for integrity using displacement and bumping techniques. Cementing operations were conducted using expandable slurry systems, with plug bumping confirmed at 3000 psi. The top of liner was tagged at 7326.5 ft, and the casing hanger was landed and packed off to isolate the annulus.

Perforation and Logging

Tubing-conveyed perforation (TCP) was used to re-establish reservoir access. Guns were deployed to target depths in both zones, with depth correlation confirmed via wireline logs (refer to figure 1). Spacer subs were used to isolate perforation intervals, enabling precise targeting and preventing cross-zone communication.

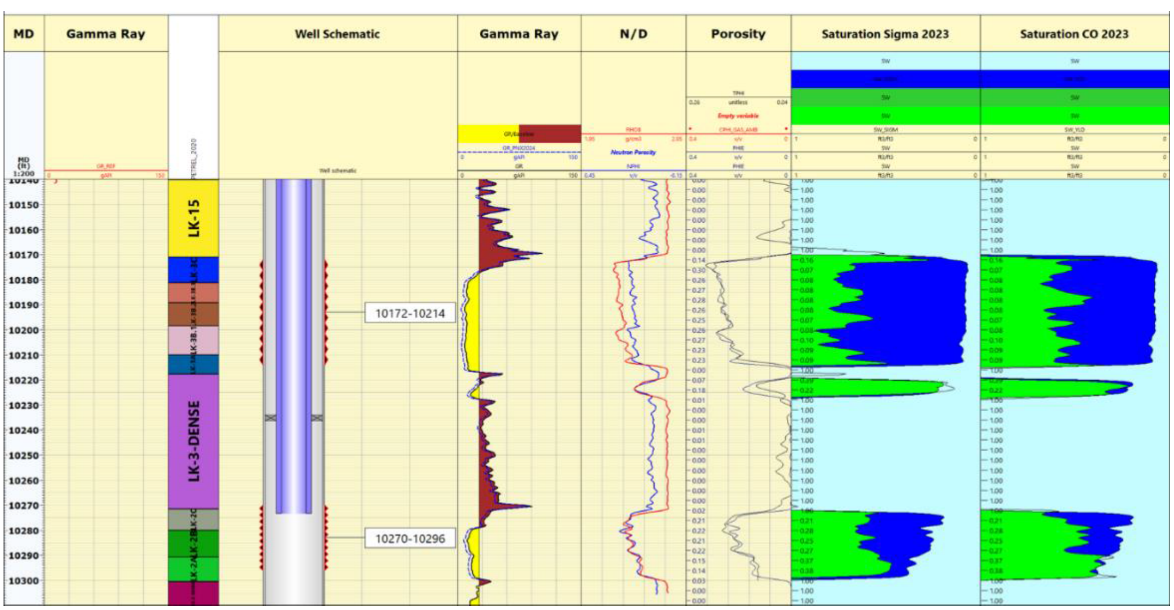

Figure 1—Wellbore selective perforations and the saturation logs

High-performance perforation charges were selected to maximize reservoir contact, offering:

- Enhanced penetration depth
- Increased flow area
- Optimized performance in stressed formations

Post-perforation, the well was circulated and cleaned, followed by logging operations including cement bond, corrosion, and gyroscopic surveys. Logging tools were deployed safely under pressure control, confirming casing integrity and cement placement.

Completion String Installation and Testing

The installation of the smart completion string was carried out in a structured and methodical sequence. Both lower and upper zone assemblies consisting of interval control valves (ICVs), solid gauge mandrels (SGMs), and packers were deployed with careful handling of pre-installed control lines and electrical cables. The upper assembly was delivered with pre-installed gauge and control line pigtails, which streamlined rig-floor operations. Packers were set at designated depths and pressure-tested to 5,000 psi for 15 minutes, with all hydraulic and electrical connections verified at each stage.

To accommodate future artificial lift requirements, four gas lift mandrels were installed, with control lines routed through bypass slots. Side pocket mandrels were included as contingency options in case of upper FCV malfunction. A safety valve was installed and function-tested to ensure emergency shut-in capability of the production string.

Smart Completion Deployment

The smart completion string was assembled with two-zone hydraulic interval control valves (ICVs), permanent downhole gauges (PDHGs), and gas lift mandrels (GLMs) (refer to figure 2, pre and post workover completion diagram). Electrical and hydraulic splicing was performed and pressure-tested to 10,000 psi. The completion was run in stages, with packers set and inflow tests conducted to validate system functionality.

Hydraulic and Electrical Integrity: All hydraulic splices were pressure-tested up to 6500 psi (externally and internally), and electrical splices were tested to 10,000 psi, ensuring robust integrity before deployment. Control lines and downhole gauges for the lower assembly were preinstalled during workshop preparation.

Both ICVs were closed in advance during wireline and slickline operations to maintain well control and prevent crossflow. The tubing hanger operation was executed without significant difficulties, and control lines were successfully routed through the tubing hanger.

Full Cycle Testing: Both upper and lower ICVs underwent full cycle tests, with hydraulic actuation and return volumes recorded for each position, confirming operational reliability. Final valve positions were set with the lower valve closed and the upper valve fully open, in accordance with operational requirements.

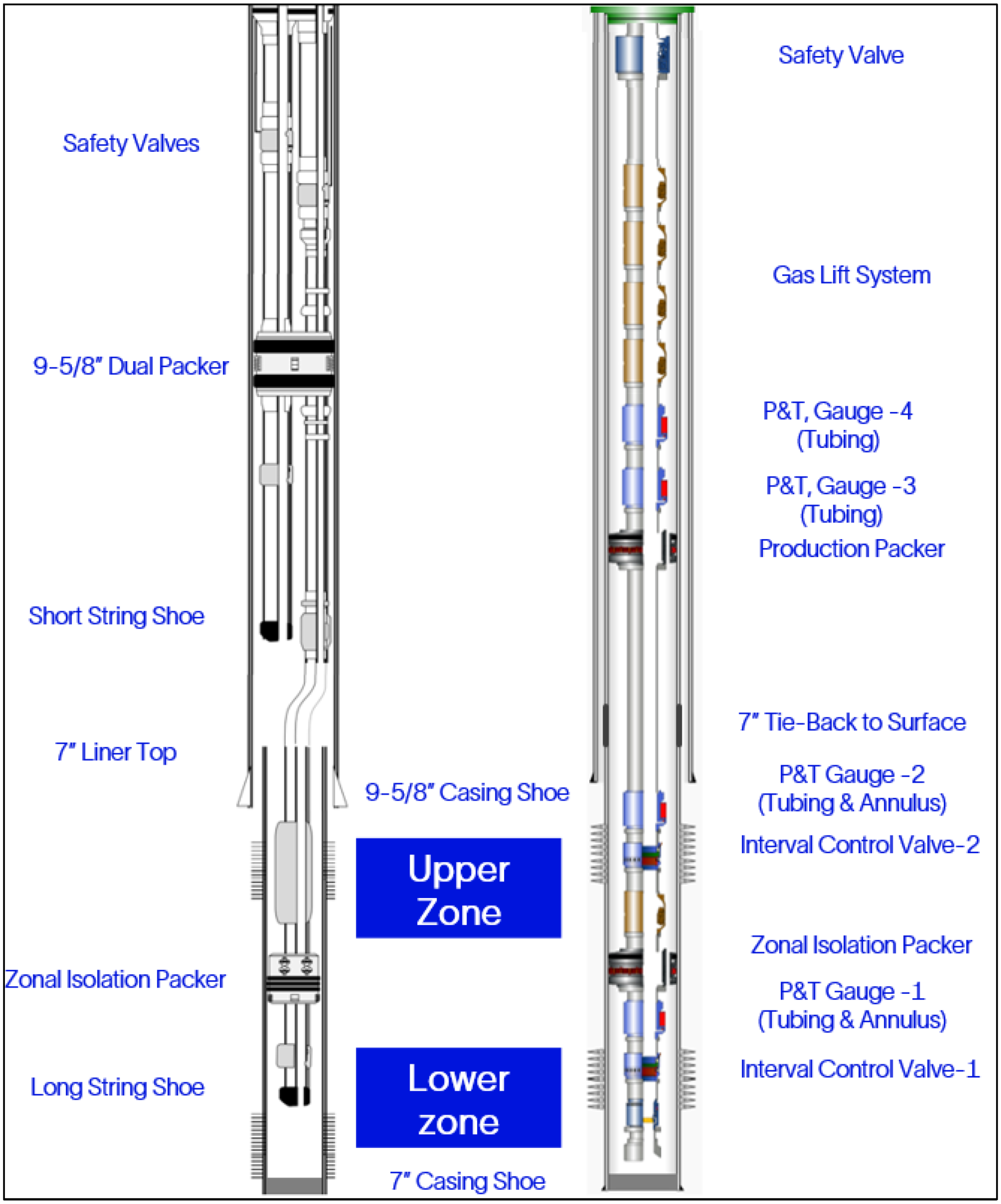

Figure 2—Pre and post workover completion diagram.

Safety Practices and Operational Discipline

Safety remained a top priority throughout the workover. Daily safety briefings, behavioral observations, and leadership safety rounds were conducted. Topics included confined space entry, working at height, and mechanical lifting. Rig crews demonstrated strong compliance with work management systems, including permit-to-work (PTW), job safety analysis (JSA), task risk assessment (TRA), and toolbox talks (TBT). Emergency drills were executed with full participation and timely response.

Well control was maintained throughout, with BOP testing and preparatory cleanout runs. All operations were executed with strict adherence to safety protocols, including scheduled safety time-outs, H_2S drills, and fire drills.

Operational Best Practices

Job Planning: Detailed engineering simulations and risk assessments ensured readiness for complex operations.

Debris Management: Effective cleanout strategies minimize risk during completion deployment.

Logging & Perforation: Accurate depth correlation and tool handling ensured successful reservoir access.

Smart Completion Integration: Seamless deployment of ICVs, PDHGs, and GLMs enabled real-time zonal control.

Safety Culture: Proactive safety management and crew engagement ensured incident-free execution.

Smart completion design and Application Highlights

The well completion downhole products serve as the interface between the reservoir, surface well control systems and production facilities. The Smart Well Completion Technology enables control and monitoring of well production or injection at the reservoir level by downhole choking and permanent down hole pressure and temperature gauges. To meet well objectives, the completion included ICVs with custom chokes, PDHGs for tubing and annulus monitoring, and zonal isolation packers. Two additional PDHGs were placed above the production packer, spaced 100 ft apart. GLMs and a TRSCSSV were installed in a 7" tie-back casing.

The ICVs used in this completion were equipped with choke modules resilient to erosive forces and customizable with different port sizes to accommodate a wide range of flow/injection rates. Final choke configurations were selected based on outcomes from advanced completion modeling studies, ensuring optimal performance under expected reservoir conditions are achievable. Choke settings allow for variable flow areas, enabling operators to fine-tune inflow rates across different operational scenarios. This flexibility supports both reservoir protection and production optimization (refer to table 1, choke size variations).

Table 1—Summarizes 3-1/2" ICV (TRFC-HDM family) choke size variation for respective positions. Final choke size selected for Project (column 4).

Engineering details	Min Flow area (in^2)	Max flow area (in^2)	Project Customized Choke sizes (in^2)	Choke Size Adjustable Yes / No
CHOKE SIZE POS. 1 (CUM AREA – SQ. IN.)	0.014	0.061	0.014	YES
CHOKE SIZE POS. 2 (CUM AREA – SQ. IN.)	0.028	0.122	0.028	YES
CHOKE SIZE POS. 3 (CUM AREA – SQ. IN.)	0.042	0.184	0.043	YES
CHOKE SIZE POS. 4 (CUM AREA – SQ. IN.)	0.056	0.367	0.063	YES
CHOKE SIZE POS. 5 (CUM AREA – SQ. IN.)	0.069	0.551	0.095	YES
CHOKE SIZE POS. 6 (CUM AREA – SQ. IN.)	0.083	0.796	0.157	YES
CHOKE SIZE POS. FULL OPEN (CUM AREA – SQ. IN.)	6.187	7.204	6.565	YES
CHOKE SIZE POS. FULL CLOSE	0	0	0	NO

Proven during technology qualifications and well installation experiences in high capacity applications, a metal-to-metal choke seal enables high unloading pressures. Unloading of pressure through the ICV is normal process during well test and well production phases thus having robust solution is required.

Digital datasets from PDHGs in combination with ICVs adjustable choke positions enabled real time inflow / outflow control, enhanced zonal isolation and inflow performance preparing the well for sustained flow without further intervention.

The ICVs utilized in this project have multi-choke positions. It has 6 choking positions and fully open and close positions. The ICV is designed to move from one to another choke position and then full open and close by applied pressure to one and then another hydraulic chamber located in the ICV. The control of ICV cycling is under the surface control system. The surface control system allows operators flawlessly shift the ICV by programmed automated surface set up (Figure 4 illustrates the ICV shifting module sketch).

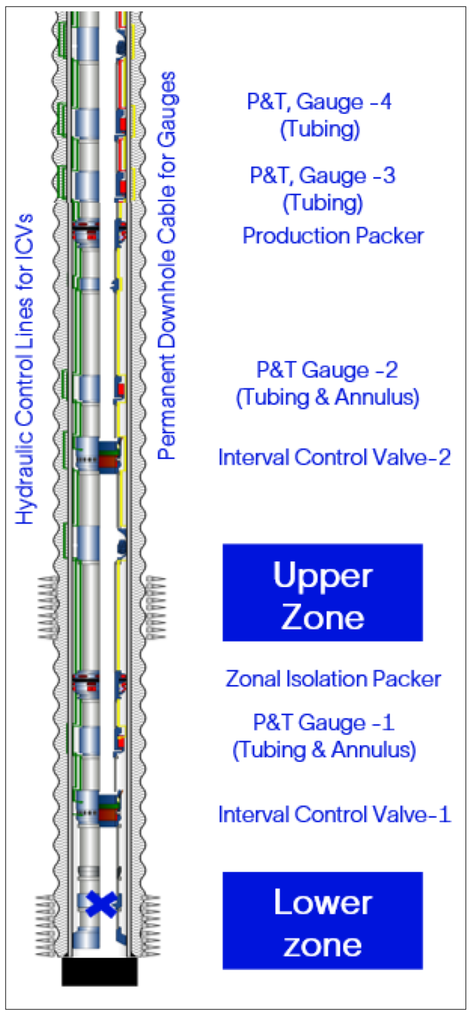

Figure 3—Illustrates well completion key components across 2 reservoir zones

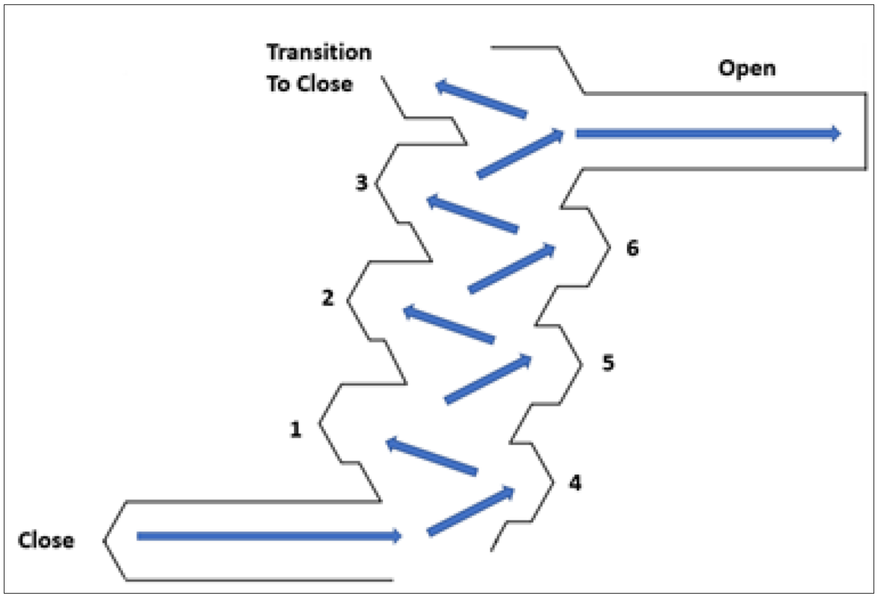

Figure 4—Illustrates 8 position ICV (TRFC-HDM family) position sequence

Well Deployment Strategy and Technology Integration

The well completion design allows production from two distinct reservoir layers in a safe manner and enables control of flow from each layer to optimize production over the life of the well. Additionally, the surface integrated panel at wellsite was equipped with hydraulic pump and a Programmable Logic Controller (PLC) to adjust valve positions automatically through a simple graphical interface. The surface set up also gathers and transmits data from downhole gauges and allows remote or local control of ICV shifting (refer to Figure 5, surface control and data delivery illustration).

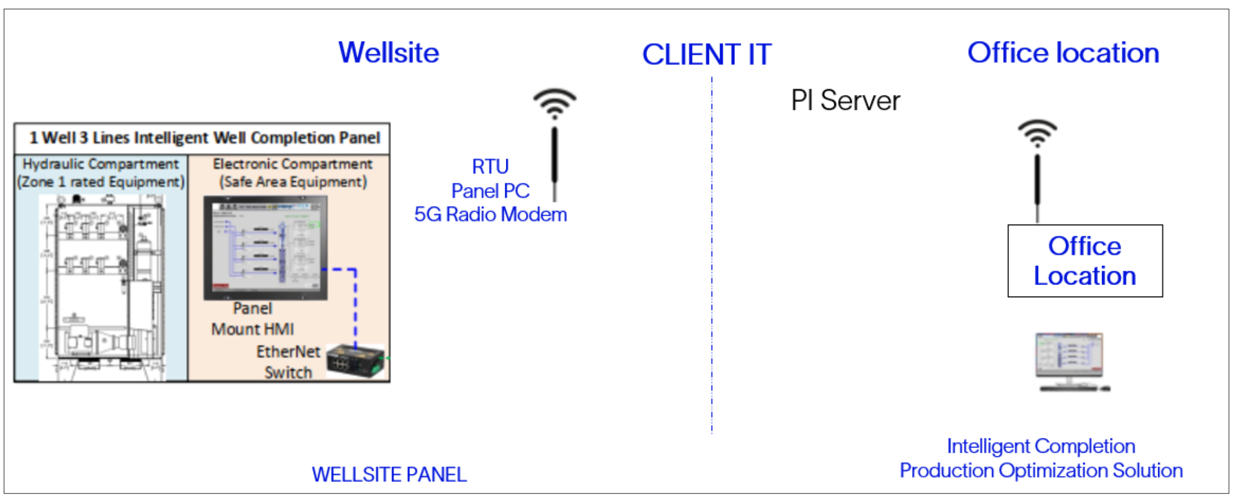

Figure 5—Illustrates well surface control and data delivery for Project.

Data from the Wellsite Panel supports performance evaluation and ICV adjustment without mechanical intervention, adapting to reservoir changes. This capability supports enhanced reservoir performance and contributes to long-term production sustainability.

Innovative Digital Solutions and Pilot Testing Outcomes

This section presents the outcomes of a pilot testing program focused on validating advanced real time digital solution for zonal production management in a dual zone well equipped with intelligent completion valves (ICVs). The testing aimed to confirm well accessibility, equipment integrity, zonal isolation, and understanding of individual zone performance using digital solution. The program used a multi-model digital back allocation algorithm to estimate zonal contributions in real time. It also provided insights into performance and water cut for each zone.

The results demonstrate how digital technologies can transform traditional well testing workflows—enhancing precision, accelerating decision-making, and unlocking new levels of operational efficiency.

Establishing Readiness and Integrity

The pilot began with a series of checks to ensure the well was fully accessible and its equipment was functioning reliably. Slickline runs were used to validate data from four downhole pressure gauges. These gauges formed the foundation of the digital monitoring system, providing continuous, high-resolution(1sec) data essential for real-time analysis.

Zonal isolation was confirmed by flowing the upper interval while keeping the lower interval closed. A clear pressure buildup in the lower zone—reaching a differential of approximately 1000 psi - demonstrated effective isolation. Intelligent valves were cycled multiple times, with each position change verified through both downhole pressure responses and surface hydraulic system checks.

Following these validations, acid stimulation was performed through individually opened ICVs. The operation resulted in a twofold increase in production, confirming both the effectiveness of the stimulation and the robustness of the completion system. Throughout this phase, digital tools played a critical role in monitoring pressure behavior, valve responses, and production trends.

Testing Individual Zonal Flow

The next phase focused on evaluating the flow performance of each interval. The upper zone was tested using a portable surface test separator and performed as expected. The digital solution enabled flow rates are validated for each choke position (refer to figure 6). However, the lower zone could not be tested separately due to its inability to flow naturally because of high water cut.

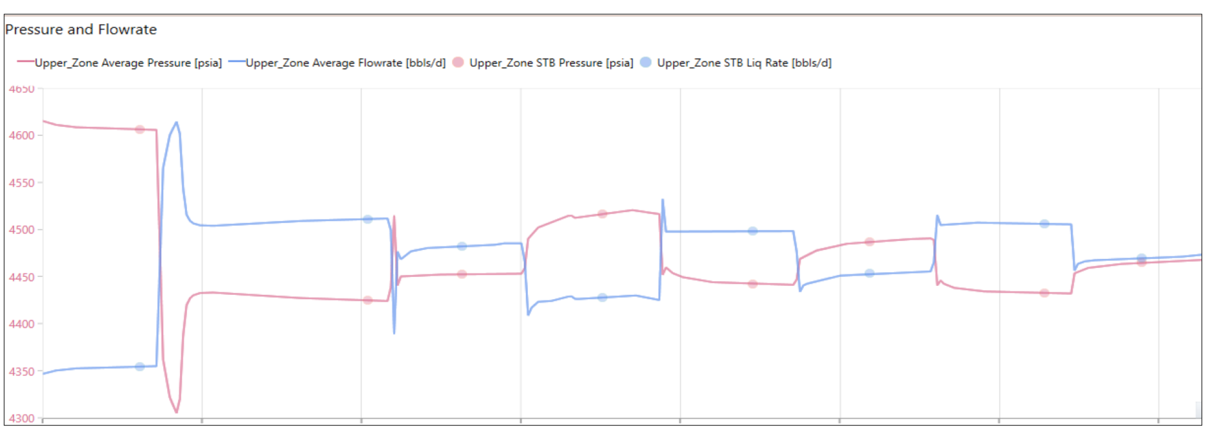

Figure 6—Targeting stable points from step rate test for IPR evaluation

To overcome this limitation, the team shifted to a digital strategy. The upper zone was fully characterized first—capturing its productivity index (PI), reservoir pressure, (refer to figure 7) water cut (WCT), and gas-oil ratio (GOR) using the automated workflows from the digital solution to process and analyze days of 1 sec data within hours. These parameters were then used as a baseline to estimate the lower zone's contribution during comingled flow using digital modeling and back allocation techniques.

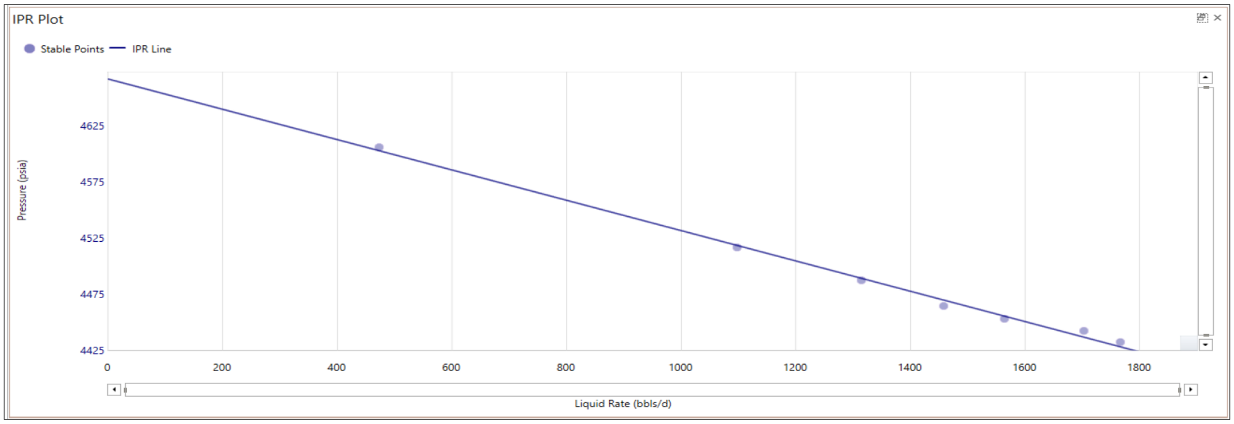

Figure 7—IPR Plot automatically evaluated for Upper zone based on digital silution

Assessing Valve Performance and Inlow Control capabilities

To validate the control capabilities of the downhole valves, the upper zone was tested across seven different ICV positions. Each adjustment produced measurable changes in surface flow rates and downhole pressure drops, confirming the valves' ability to modulate flow effectively (refer to table 2).

Table 2—Summarizes choking performance for different ICV positions.

Choke	DP,psi	Drawdown, psi
1.00	1473	158
2.00	1019	186
3.00	651	209
4.00	444	224
5.00	257	235
6.00	129	244

This testing was enhanced by digital control systems that allowed precise valve actuation and immediate feedback. The integration of surface and downhole data into a unified digital platform enabled seamless tracking of valve behavior and its impact on zonal performance.

Digital Evaluation of Zonal Performance

Digital solutions were central to evaluating the upper zone's inflow characteristics. Step-rate testing at various surface choke settings allowed the team to build an inflow performance relationship (IPR) curve. Stable flow points were used to estimate reservoir pressure (P*) and drawdown, which was then validated against historical shut-in data. Surface measurements of WCT and GOR were integrated into a real-time zonal rate calculation algorithm. This digital workflow provided dynamic insights into the zone's behavior, enabling continuous performance monitoring and rapid recalibration when needed. The digital system ensured that all data—from surface rates to downhole pressures—was processed within a consistent analytical framework. This eliminated manual interpretation errors and allowed for repeatable, high-confidence evaluations.

Digital Back Allocation: A Multi-Model Approach

One of the most innovative aspects of the pilot was the implementation of a digital back allocation algorithm. This algorithm, developed and refined through previous field applications, enabled accurate estimation of zonal contributions even under complex flow conditions.

The algorithm operates through several interconnected steps:

- **Surface Rate Estimation**: Using metering data to determine total flow.
- **Zonal Characterization:** Estimating PI, reservoir pressure, WCT, and GOR through step-rate testing.
- **Gradient Analysis:** Calculating pressure gradients from downhole sensors.
- **Hydrostatic Modeling:** Replicating actual flow conditions using fluid properties and well geometry.
- **Choke Flow Calibration:** Adjusting flow coefficients (Cv) based on real-time valve behavior and pressure data.

Each zone's flow rate was estimated using three independent models:

1. Based on IPR and drawdown.
2. Derived from the full well hydrostatic model.
3. Calculated using the calibrated choke model.

These models were continuously cross-validated using pressure gradient checks (refer to figure 8). Any inconsistencies in GOR or WCT estimates were flagged through gradient mismatches, ensuring high accuracy and reliability. The digital platform's ability to reconcile these models in real time was a key enabler of operational success.

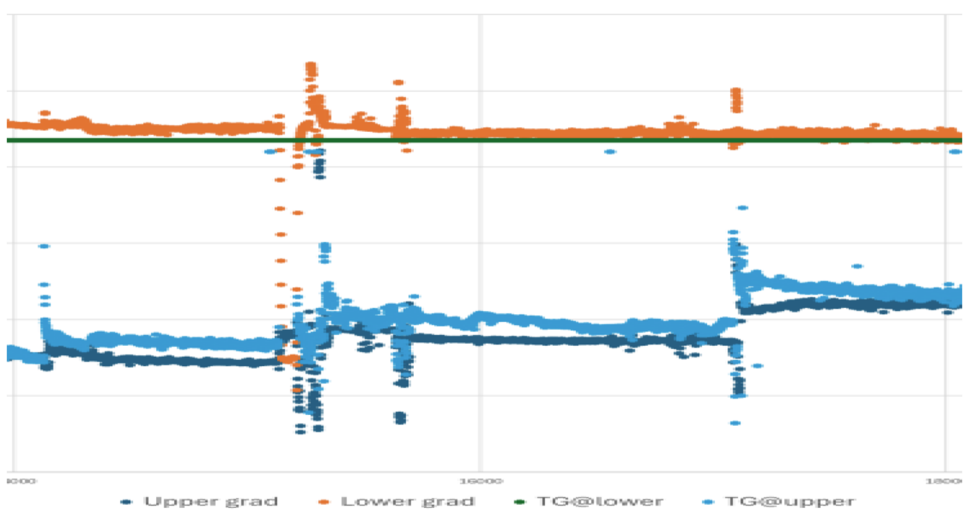

Figure 8—Gradient check for Upper & Loewer intervals (calculated - dark, actual measured - lighter)

Transition to Comingled Flow

With the upper zone fully characterized, the well was transitioned to comingled production. The digital system provided validated estimates for the upper zone:

- P* ≈ 47XX psi
- PI ≈ 7.X bbl/d/psi

- WCT ≈ 2%
- GOR ≈ 1XXX
- Gradient ≈ 0.24X psi/ft

These values were confirmed through all three models and surface flow metering. The digital solution ensured that each model remained calibrated and aligned with live data, maintaining consistency between predicted and observed performance.

Ten combinations of ICV positions were tested during the comingled flow. Surface stabilization and PLT assessments were conducted (refer to table 3). Although the lower zone could not be tested individually due to persistent high water cut, its contribution was estimated using the digital back allocation algorithm.

Table 3—Zonal contributions based on performed different zonal ICV combinations

Position (U/L)	4 /7	4/6	2/6	3/6	3/5
Upper Zone (**Liquid %)**	52.3	56.5	40.8	50.8	52.2
	65.3	67.5	44.4	52.8	57
Lower Zone (**Liquid %)**	47.6	43.5	59.2	49.2	47.8
	34.7	32.5	55.6	47.2	43

Estimating Lower Zone Performance

Using the validated upper zone parameters as a reference, the digital system estimated the lower zone's performance during comingled flow:

- P* ≈ 48XX psi
- PI ≈ 3.X bbl/d/psi
- WCT ≈ 7X%
- GOR ≈ 8XX
- Gradient ≈ 0.28X psi/ft

These estimates were validated through surface measurements and PLT runs (refer to figure 9). The alignment of pressure gradients across vertical sections confirmed the accuracy of the digital model and the reliability of the back allocation process.

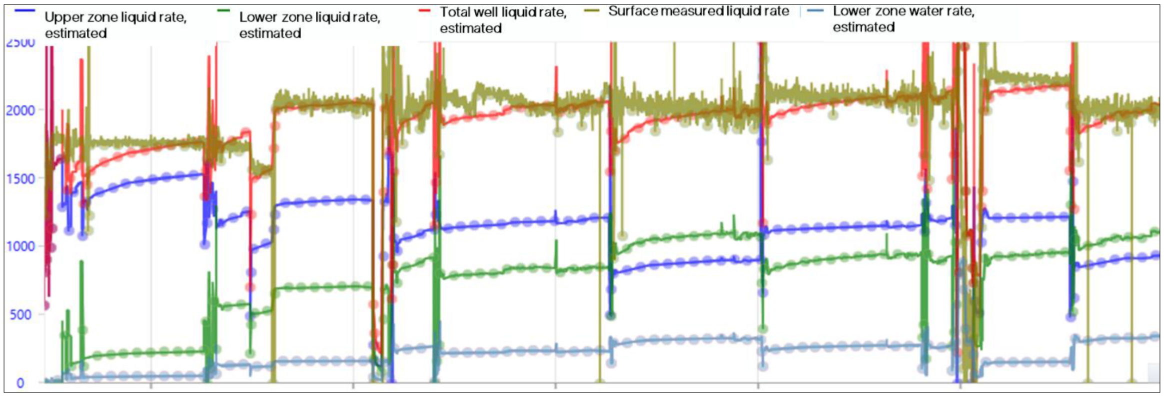

Figure 9—Final estimated Zonal liquid rates and water rates during comingled flow

The ability to derive meaningful insights from indirect data sources enabled by digital modeling was a major breakthrough in this pilot.

Digital Solution Validation and Best Practices

The digital solution used in this pilot has been successfully deployed in various regions and reservoir types (refer to Fig 10). Its modular architecture allows for customization based on well design, fluid properties, and operational constraints.

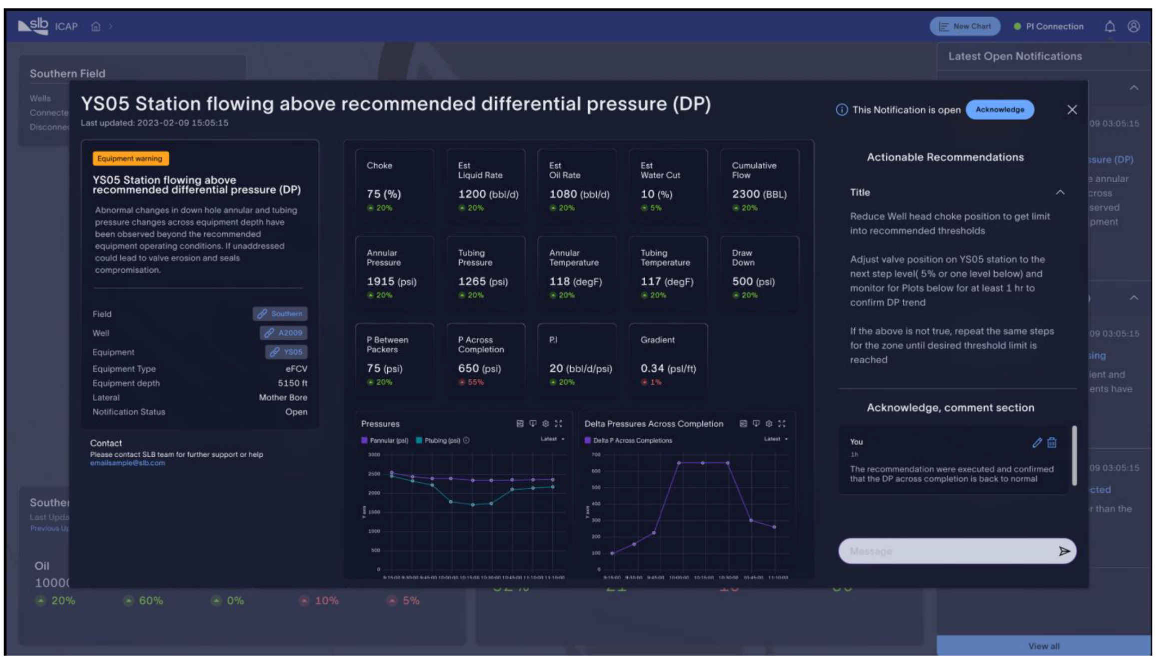

Figure 10—Digital solution reference and screen outlook.

To ensure long-term sustainability and accuracy, the following best practices are recommended:

- Continuous monitoring of pressure and temperature trends through events and alarm notifications
- Periodic surface separator measurements when significant changes of well conditions are noticed
- Choke and hydrostatic model recalibration when significant changes of well conditions are noticed
- Cross-validation with PI-based models

- Gradient consistency checks

These practices help maintain model integrity and ensure that digital predictions remain aligned with field realities.

Key Outcomes and Impact

The pilot testing confirmed the effectiveness of digital solutions in managing and understanding zonal production in commingled production well without intervention. It also demonstrated how production engineers can benefit from automatic processing of millions of raw data points into insights and be able to make decisions on the fly to optimize production and maximize reservoir performance. The well was successfully tested and ready to comingle flow, with validated zonal contributions and an incremental inflow increase of approximately 25% compared to untreated conditions.

More importantly, the digital system together with Inflow control system enabled a level of precision, responsiveness, and insight that traditional methods could not deliver. By integrating real-time data, predictive modeling, and automated validation, the digital workflow transformed the way zonal performance was understood and optimized.

This case study highlights the growing importance of digital technologies in modern reservoir management (refer to figure 11). As wells become more complex and data-rich, digital solutions will be essential for unlocking their full potential.

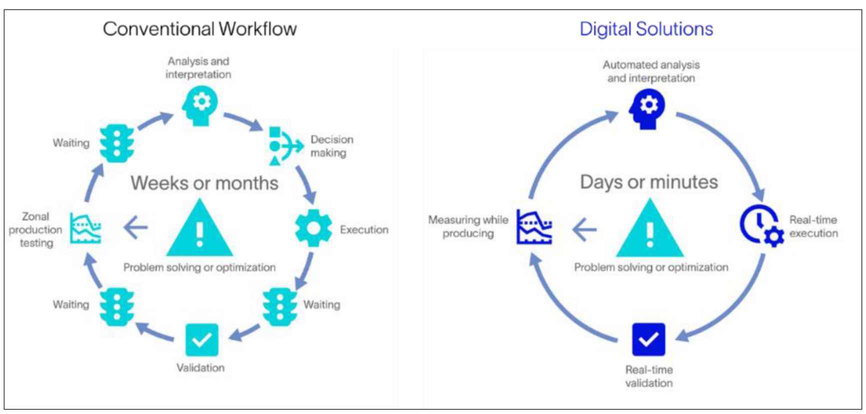

Figure 11—Comparison between conventional workflow and digital solutions applied for candidate well.

Results Discussion

The project outcome stands as a benchmark for future similar intelligent well interventions in mature fields. By integrating advanced completion technologies with digital solutions, rigorous safety practices, and collaborative execution, the operation restored production capability and extended the well's lifecycle. The insights gained from this project offer valuable guidance for future smart completion deployments and complex workover scenarios.

The rehabilitation of candidate well through smart completion deployment has delivered measurable improvements in both well performance and reservoir management. After integrity and flow issues, the well was recompleted with a dual-zone intelligent system using ICVs, PDHGs, GLMs, and a digital solution. This configuration enabled selective and commingled production from both target reservoir zones, restoring the well's functionality and unlocking previously stranded reserves.

Post-completion results confirmed stable production from upper reservoir zone at approximately 1,000 BOPD with minimal water cut (<2%), while the lower reservoir zone contributed an additional 300–400 BOPD through commingled flow, despite its high water cut (~70%). The commingled strategy allowed the upper reservoir zone to act as a natural lift mechanism for the lower reservoir zone, validating the engineered flow design and eliminating the need for artificial lift systems. During well testing, the upper zone ICV was cycled a total of 105 times, and the lower zone ICV was cycled 72 times to identify optimal flowing positions. These cycles were instrumental in fine-tuning inflow control and validating zonal contribution strategies.

The completion string was built with corrosion-resistant alloys and contingency features such as wireline retrievable plugs and side pocket mandrels. All hydraulic and electrical splices were pressure-tested to 6,500 psi, ensuring integrity under high-stress conditions. Dual-zone sensors enabled real-time monitoring, with data streamed to surface systems for continuous control. This enabled dynamic inflow management and zonal isolation, improving sweep efficiency and reservoir conformance.

Operationally, the intervention avoided the need for a replacement well and eliminated future rigless workovers for selective production. It also reduced reliance on wireline gradient surveys and PLTs for zonal allocation, resulting in an estimated capital savings of $4.8 million.

A tailored reservoir monitoring plan was implemented, including monthly production tests, multi-rate testing, and annual PLTs. These activities supported model calibration and production allocation workflows for the validation of the digital solution, which showed strong alignment with actual well performance. Leveraging digital solution capabilities, the objective is to reduce the operational expenditure through optimized reservoir monitoring plan.

This case demonstrates the technical feasibility and operational value of deploying smart completions in complex well environments. It highlights the importance of integrated planning, rigorous simulation, and adaptive execution in restoring production and extending well life. The learnings from the project provides a practical framework for future smart well deployments in similar reservoir settings.

Conclusion

This case study demonstrates the transformative impact of integrating smart completion systems with advanced digital solutions in a mature, integrity-challenged well. By deploying dual-zone hydraulic ICVs, PDHGs, and a programmable surface control system, ADNOC Onshore successfully restored production from both reservoir zones—achieving selective and commingled flow without the need for drilling a replacement well. The digital back allocation algorithm enabled real-time zonal performance estimation, even under complex flow conditions, while rigorous planning and execution ensured operational safety and system integrity. The project not only delivered a 25% increase in inflow and $4.8 million in CAPEX savings but also set a new benchmark for intelligent well rehabilitation in the region.

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